

Experiment :- 1

AIM :- What is SDLC. Explain all the SDLC Models.

SDLC (Software Development Life Cycle) :-

Software Development Life Cycle is a process followed by software industries to design, develop, test, deploy, and maintain software applications.

SDLC helps software developers, project managers, and organizations plan, design, develop, test, and maintain software effectively.

SDLC Phases :-

1. Requirement Gathering
2. Planning
3. Design
4. Development (Coding)
5. Testing
6. Deployment
7. Maintenance

SDLC Models :-

1) Waterfall Model :-



The Waterfall Model is a software development model in which development activities are completed in a sequential order such as requirement analysis, design, implementation, testing, deployment, and maintenance.

i) Requirement Analysis

In this phase, all the requirements of the customer are gathered and documented.

Activities:

- Understanding customer needs
- Requirement collection
- Preparing SRS document

ii) System Design

The system architecture and design are prepared based on the requirements.

Activities:

- Database design
- User interface design
- System architecture creation

iii) Implementation (Coding)

In this phase, developers write the actual source code according to the design specifications.

Activities:

- Coding
- Program development
- Module creation

iv) Testing

The developed software is tested to identify bugs and ensure that it meets user requirements.

Types of Testing:

- Unit Testing
- Integration Testing
- System Testing

v) Deployment

After successful testing, the software is released to the customer or end users.

Activities:

- Installation
- Configuration
- User training

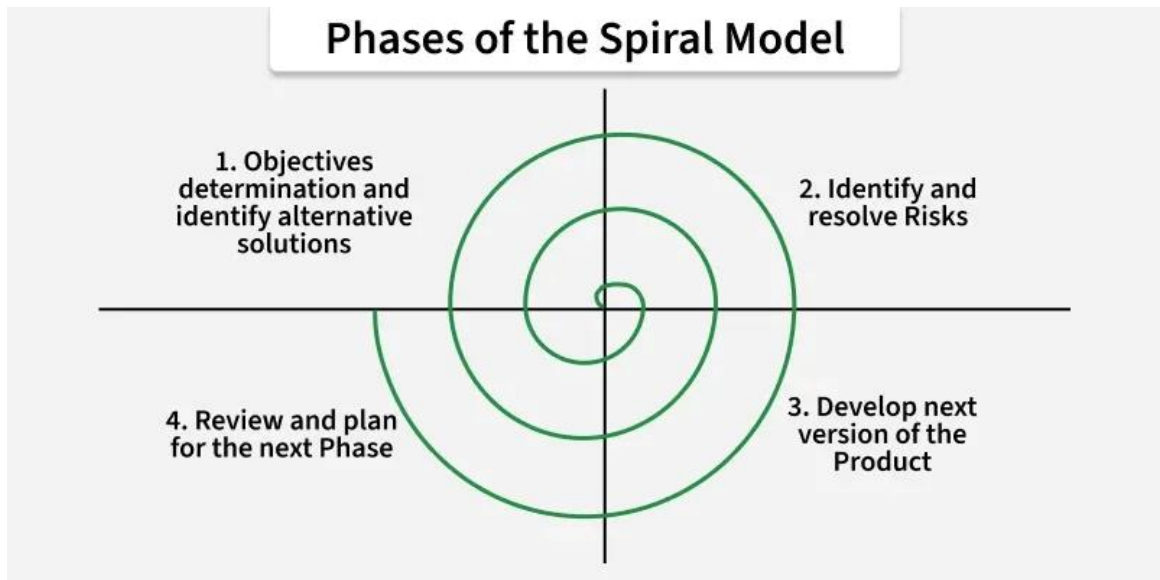
vi) Maintenance

After deployment, maintenance is performed to fix issues and improve the software.

Activities:

- Bug fixing
- Performance improvement
- Software updates

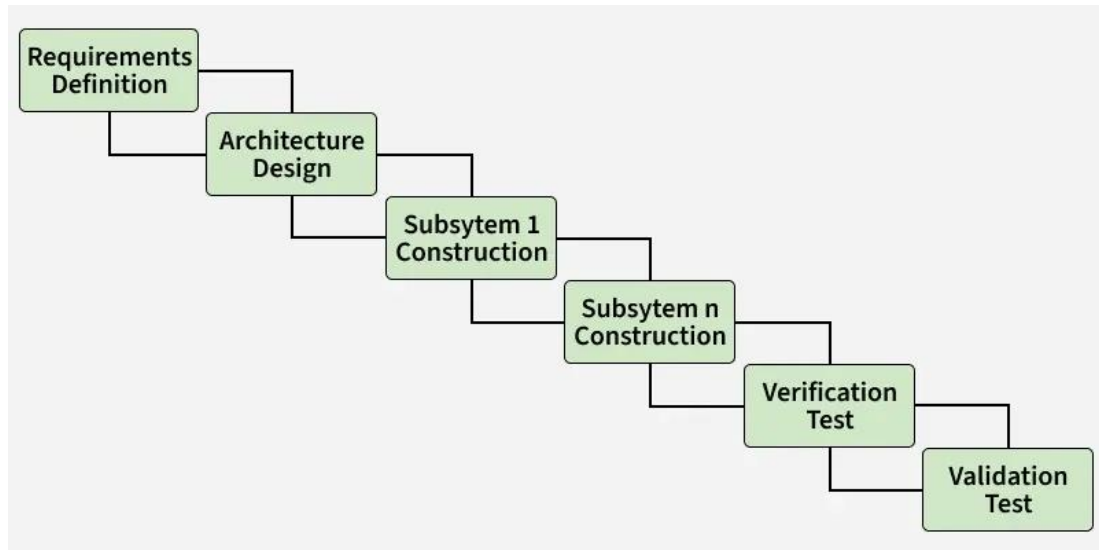
2) Spiral Model:-



The Spiral Model is an SDLC model in which the development process passes through repeated phases of planning, risk analysis, engineering, and evaluation until the software is completed.

- i) **Planning**
Requirements are gathered and objectives are identified.
- ii) **Risk Analysis**
Possible risks are identified and solutions are planned.
- iii) **Engineering**
Coding and development of the software take place.
- iv) **Evaluation**
The customer evaluates the developed software and gives feedback.

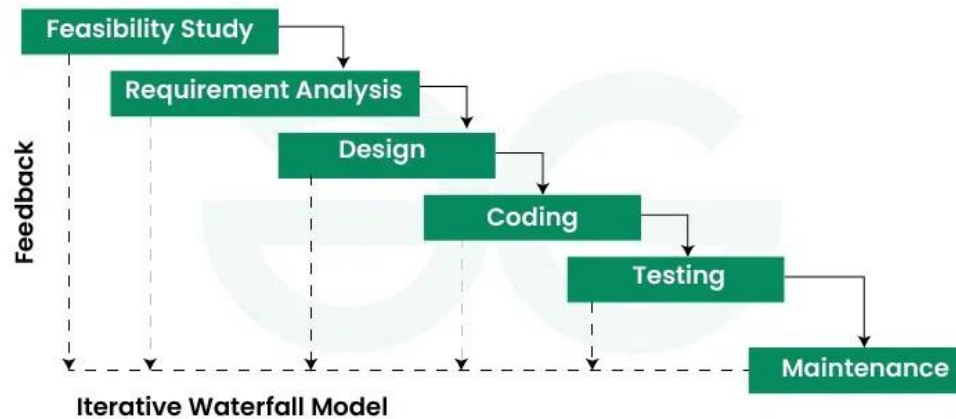
3) Incremental Model :-



The Incremental Model is an SDLC model in which software is developed and delivered in smaller functional parts.

- i) **Requirement Analysis**
Requirements are collected and divided into increments.
- ii) **Design**
Design is prepared for each increment.
- iii) **Development**
Coding is done for a specific increment.
- iv) **Testing**
Each increment is tested separately.
- v) **Deployment**
The increment is delivered to users.

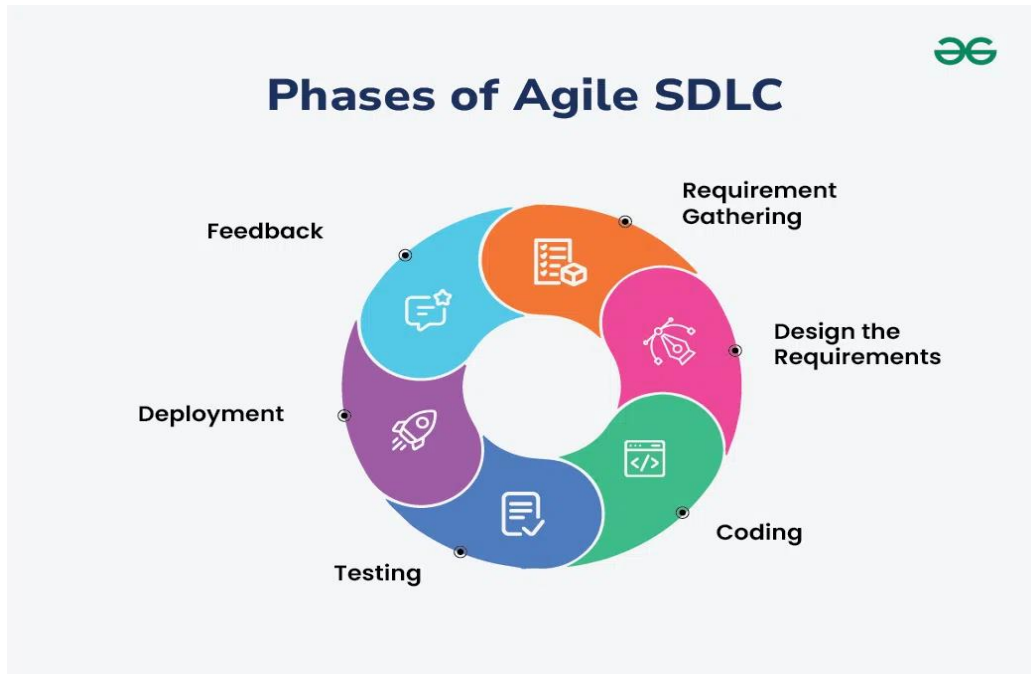
4) Iterative Model :-



The Iterative Model is an SDLC model where software is developed repeatedly by refining and improving it in every cycle.

- i) **Requirement Gathering**
Basic requirements are collected.
- ii) **Design**
Initial design is prepared.
- iii) **Development**
Software is developed in iterations.
- iv) **Testing**
Testing is performed after each iteration.
- v) **Review**
Feedback is collected and improvements are made.

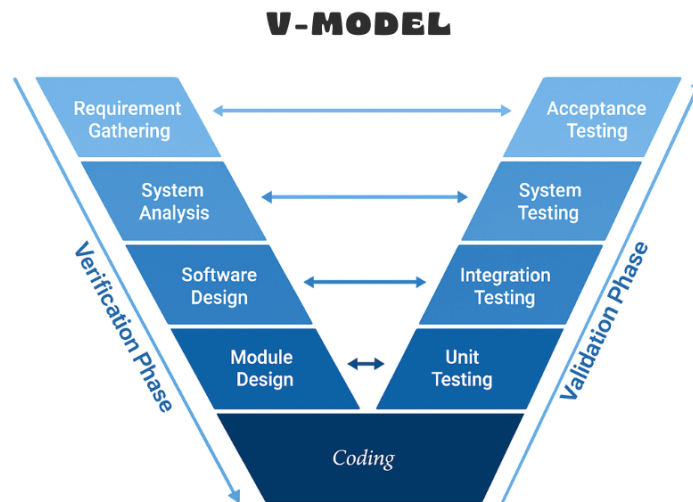
5) Agile Model :-



The Agile Model is an iterative and incremental software development approach that delivers software quickly with continuous customer feedback.

- i) **Requirement Gathering**
Requirements are collected from customers.
- ii) **Planning**
Tasks are divided into small sprints.
- iii) **Development**
Coding is done for each sprint.
- iv) **Testing**
Testing is performed continuously.
- v) **Release**
Software is released incrementally.
- vi) **Feedback**
Customer feedback is collected for improvement.

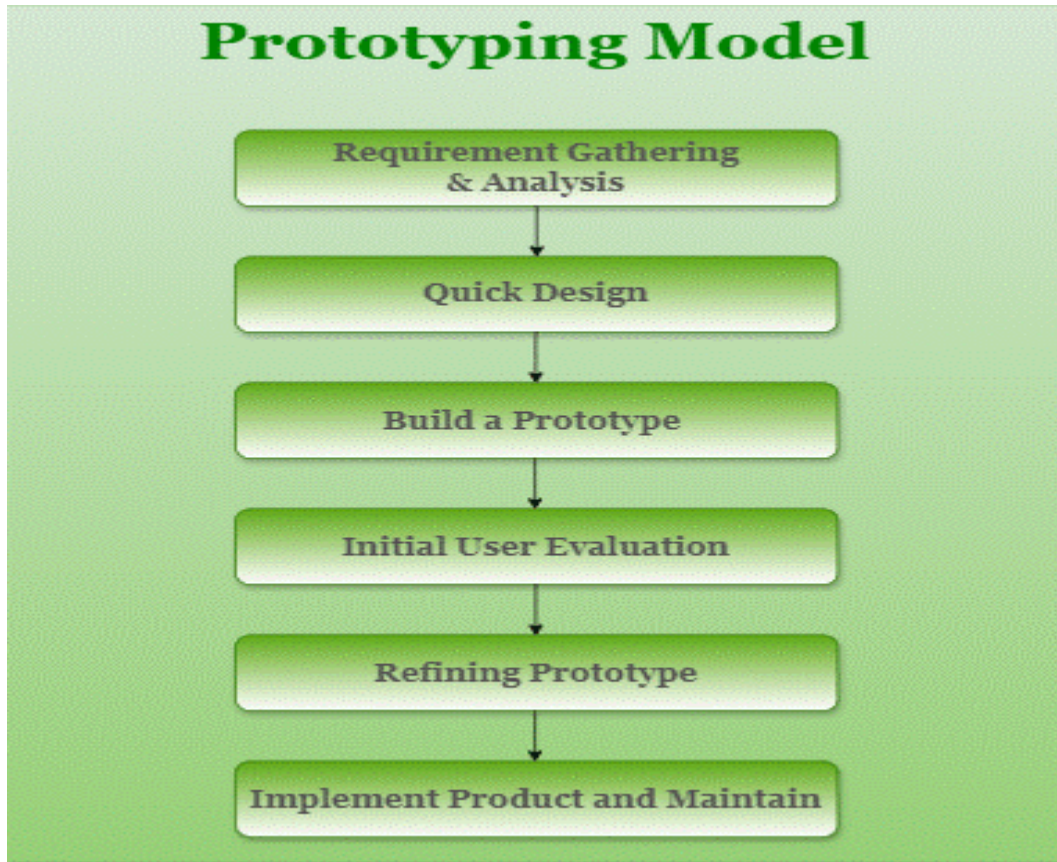
6) V-Model :-



The V-Model is an SDLC model where verification and validation activities are performed simultaneously during software development.

- i) **Requirement Analysis**
Customer requirements are collected and analyzed.
- ii) **System Design**
Overall system architecture and design are prepared.
- iii) **Architecture Design**
The system is divided into different modules.
- iv) **Module Design**
Detailed design of each module is created.
- v) **Coding**
Developers write the actual source code.
- vi) **Unit Testing**
Individual modules are tested separately.
- vii) **Integration Testing**
Combined modules are tested together.
- viii) **System Testing**
The complete software system is tested.
- ix) **Acceptance Testing**
The customer verifies and approves the software.

7) Prototype Model :-



The Prototype Model is an SDLC model in which a sample version of the software is developed first to gather user feedback.

- x) **Requirement Gathering**
Basic requirements are collected.
- xi) **Quick Design**
A simple design is prepared.
- xii) **Prototype Development**
A working prototype is developed.
- xiii) **User Evaluation**
Users test the prototype and provide feedback.
- xiv) **Refinement**
The prototype is improved based on feedback.
- xv) **Final Development**
The actual software is developed.